

A High-Refresh Rate and Large-Size OLED Display with DRD Gate Driver for Premium TVs

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Abstract

This paper presents a novel high-refresh rate and large-size OLED display for premium TVs using a double-rate driving (DRD) method, dual-gate pixel architecture, and small-area gate driver in panel (GIP). These technologies halve the number of source driver ICs, enable stable 165 Hz operation, achieve a pixel charging ratio of over 80% across all gray levels, reduce the gate driver design area by approximately 50%, and minimize panel bezel width. The proposed architecture was successfully implemented in a 65-inch 4K OLED panel without additional mask processes, demonstrating strong feasibility for next-generation premium OLED TVs.

Author Keywords

High-refresh rate OLED display; double-rate driving; dual-gate pixel; gate driver in panel; premium TV.

1. Introduction

Organic light-emitting diode (OLED) displays have become a leading technology in the premium TV market due to their superior image quality, high contrast ratio, fast response time, and energy efficiency [1, 2]. High-resolution and large-size OLED panels, in particular, offer an immersive viewing experience and are essential for premium products that demand both advanced performance and sophisticated design [3]. Significant technical progress has been made in large-size and high-resolution OLED displays. In the case of the TFT backplane, Oxide TFTs have been used as OLED display backplanes, which are different from a-Si TFTs commonly used for LCDs. Its structure has been developed into a coplanar type with small and low parasitic capacitance to increase the pixel aperture ratio [4].

However, realizing high-resolution and large-size OLED panels introduces several technical challenges. First, the increasing number of data lines in large-size panels requires a large number of data driver ICs, leading to higher system cost and increased border circuit complexity [5, 6]. To solve this problem, more efficient driving methods are necessary. Second, minimizing panel bezel width is essential for premium TV applications, yet conventional gate driver-in-panel (GIP) circuits occupy a significant portion of the bezel area [7].

While double-rate driving (DRD) methods with dual-gate pixel architectures have been investigated and commercialized in LCD displays [8, 9], these methods reduce the number of data driver ICs by 50% versus single-rate driving (SRD) with single-gate pixel, enabling cost-effective solutions. Yet, their halved gate-scanning time exacerbates pixel charging limitations, restricting refresh rate of LCD panels. The line time available for charging a pixel also decreases in inverse proportion to the number of horizontal gate lines, as shown in Figure 1. In the case of the OLED display with 4K resolution, the gate line time is 1.3 μ s when operating at 165 Hz. Also, applying DRD increases the total number of gate lines, which can potentially enlarge the panel bezel when conventional GIP circuits are used.

In this work, we propose a novel OLED panel architecture combining double-rate driving (DRD), a dual-gate pixel structure, and a small-area gate driver in panel (GIP). The proposed technologies halve the number of data driver ICs, enable stable 165 Hz operation, and significantly reduce bezel width. These technologies were successfully implemented in a 65-inch 4K OLED panel, demonstrating strong feasibility for next-generation premium OLED TVs.

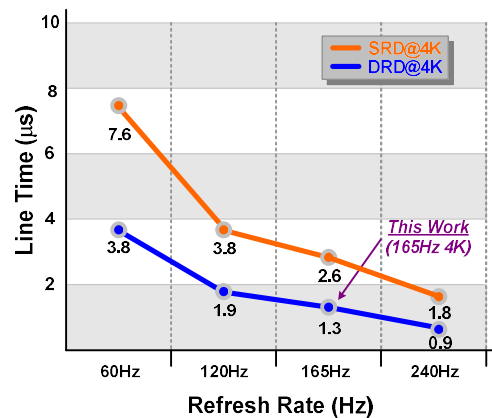


Figure 1. Line time of the 4K resolution display panel at various frame rates

2. Novel DRD OLED Display Technology

2.1 Pixel Structure and Driving Method

Conventional single-rate driving (SRD) requires one data update per line within one horizontal period, forcing OLED panels to employ a large number of source output channels. To address this limitation, we introduce a double-rate driving (DRD) method in which pixel data are updated twice within a single horizontal period. This approach effectively doubles data throughput and enables a significant reduction in the number of source driver ICs required for the same resolution panel.

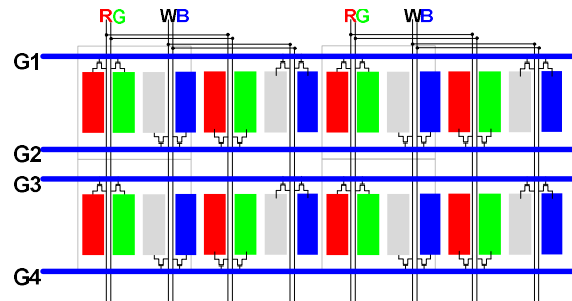


Figure 2. Dual gate pixel architecture optimized for DRD operation

Figure 2 shows the proposed dual-gate pixel structure of OLED panels. The dual gate pixel structure designed for the DRD method is specifically optimized to enhance the performance and efficiency of large-size OLED panels. This structure incorporates two gate lines per pixel, enabling the simultaneous or alternating activation of pixel circuits to achieve 165 Hz high-refresh rate operation and improved signal processing. In this configuration, thin-film transistors (TFTs) are strategically arranged to support the dual gate driving mechanism. The dual gate lines are positioned to minimize signal delay and ensure uniform electrical characteristics across the panel. Experimental verification confirms that the dual-gate pixel structure supports stable DRD operation while maintaining uniform image quality. This innovative design contributes to the advancement of large-area OLED panel technology, offering improved performance, reduced manufacturing complexity, and enhanced competitiveness in the premium TV market.

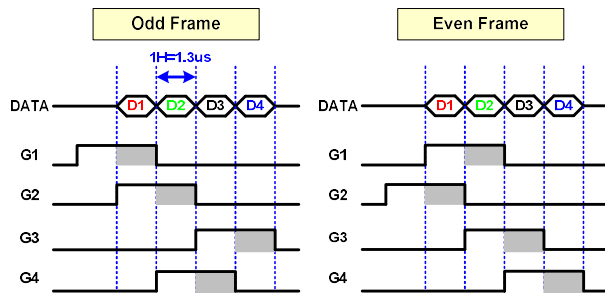


Figure 3. Gate and Data Timing of high-refresh rate DRD Methods

The proposed concept of the double rate driving method with gate and data timing is shown in Figure 3. In this method, the output sequence alternates between frames, allowing for enhancing pixel charging rate and reducing power consumption. The gate driver output waveform diagram illustrates the timing and amplitude of the signals generated during this process. It visually represents how the signals are modulated and synchronized to achieve the desired operational characteristics, ensuring improved display quality and system reliability.

2.2 Small-Area DRD Gate Driver in Panel

A proposed multi-function gate driver for the DRD driving method is shown in Figure 4. It consists of a circuit block for normal driving and a circuit block for high-speed driving. By selecting control signals, sequential output is generated in the normal operation mode, whereas dual-gate pixel DRD outputs are generated simultaneously in the high-speed driving mode.

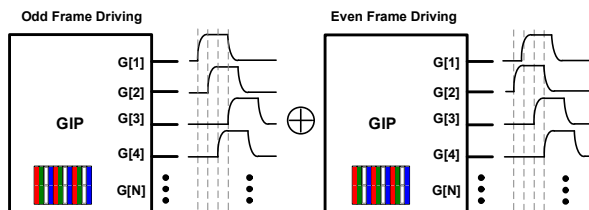


Figure 4. DRD driving concept and integrated multi-function gate driver waveforms

Applying the DRD method increases the total number of gate lines, which can potentially enlarge the panel bezel when conventional GIP circuits are used. To overcome this limitation, we developed a

compact GIP architecture featuring shared control logic, a reduced number of TFTs in the shift register, and an optimized output-stage configuration.

Figure 5(a) shows the conventional gate driver circuit consisting of control logic, clock signals, shift registers, and pull-up/down buffers. Figure 5(b) shows the proposed compact DRD gate driver, which minimizes the number of TFTs in the shift register and arranges pull TFTs in an even-odd alternating pattern. As a result, bezel expansion caused by the increased gate-line count in DRD operation is effectively suppressed. These improvements significantly reduce the GIP footprint while maintaining stable operation.

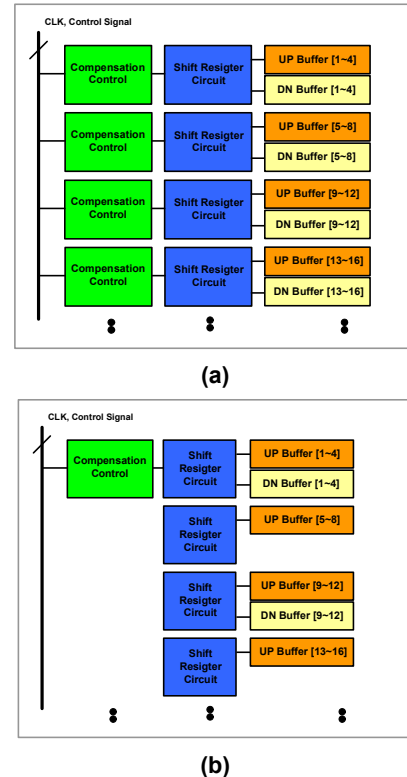


Figure 5. (a) Conventional gate driver and (b) the proposed small area DRD gate driver

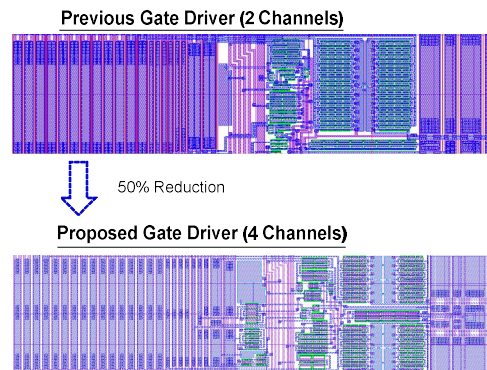


Figure 6. Comparison results of the layout design between the proposed gate driver and our previous circuit applied to the DRD 65-inch 4K OLED panel

Figure 6 compares the layout design between the proposed gate driver and the previous circuit used in conventional 65-inch OLED panels. The proposed circuit reduces the gate driver design area by approximately 50%, directly contributing to bezel reduction while maintaining stable electrical performance.

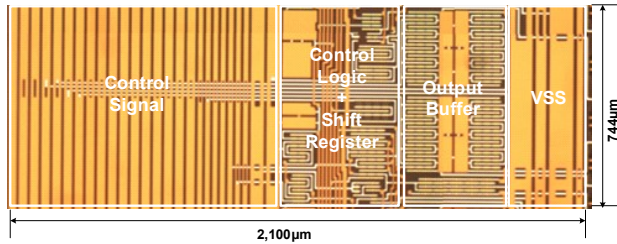


Figure 7. Optical microscope image of the fabricated gate driver applied to 31.5-inch OLED display panels

Figure 7 shows the optical microscope image of the fabricated gate driver applied to 65-inch 4K OLED display panels. It consists of the control signals, control logic circuit, shifter register, and output buffer stages, and has a structure in which multiple output buffers are connected to one shifter register circuit. The size of the proposed gate driver is 50 % smaller than that of our previous gate driver. Therefore, we reduced the size of the panel bezel by 33 %.

2.3 The OLED Panel and DRD Circuit

The proposed DRD panel and circuit was implemented on a 65-inch 4K OLED TV panel employing an oxide TFT backplane and a four-stack WOLED emission structure. The integration of DRD, the dual-gate pixel architecture, and the small-area GIP block was achieved without requiring additional mask processes or changes to OLED materials, demonstrating strong manufacturing feasibility in practical production.

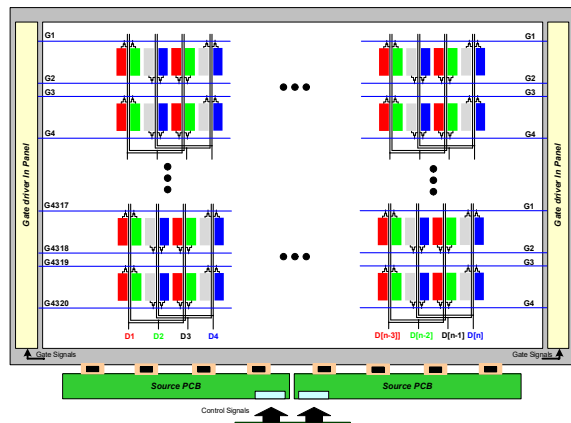


Figure 8. DRD panel and circuit platform for the 4K OLED TVs

Figure 8 illustrates the implemented OLED panel and circuit configuration of the proposed DRD architecture. The panel employs a dual-gate pixel structure optimized for double-rate driving, enabling stable operation at 165 Hz while maintaining uniform electrical characteristics across the panel.

By designing the panel based on the DRD scheme, the total number of GIP channels increased to 4,320 to support the dual-gate

configuration. Despite the increased gate-line count, the proposed compact GIP structure effectively prevents bezel expansion. In addition, the DRD architecture reduces the required number of source driver ICs from 16 in the conventional design to 8, significantly lowering system complexity and cost while maintaining stable high-speed operation.

3. Experimental Results and Discussion

The fabricated 65-inch OLED prototype demonstrates significant system-level improvements. By applying the DRD method, the number of source driver ICs was reduced by half from sixteen to eight devices, lowering system cost and border circuitry area. Stable operation at 165 Hz was achieved with no degradation in signal integrity or panel performance.

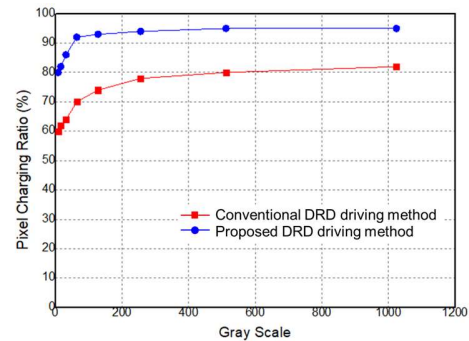


Figure 9. Measurement results of the pixel charging ratio with the proposed DRD method

Figure 9 shows the measured pixel charging ratio using the proposed DRD method with over-driving and under-driving techniques applied to enhance luminance uniformity. The charging ratio exceeded 80% across all gray levels. No degradation was observed in motion artifacts, flicker, mura, or luminance uniformity during high-refresh-rate operation.

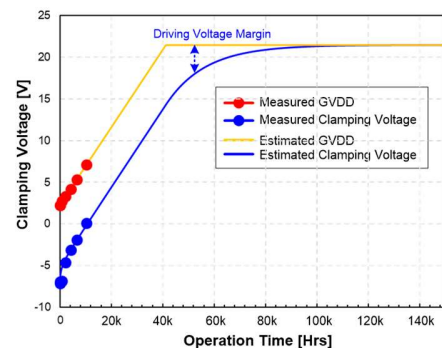


Figure 10. Measured and estimated clamping voltage at 60 °C acceleration lifetime test

Figure 10 shows the lifetime estimation results of the 60 °C acceleration lifetime test. We use clamping voltage, the lowest functioning voltage of the VDD line, to estimate the VTH of pull-down TFTs. Lowering the VDD voltage has the same effect as the positive VTH shift of pull-down TFTs, and we can therefore estimate a voltage margin before malfunction. Based on the accelerated evaluation results, the lifetime of the proposed gate driver in panels was predicted to be over 100,000 hours.

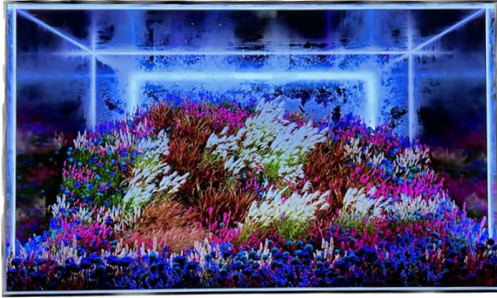


Figure 11. Photograph of the developed 165 Hz 65-inch DRD 4K OLED display

Figure 11 is a photograph of the developed 65-inch 4K OLED display panel employing an oxide TFT backplane and 4-stack WOLED. Those technologies integrate double-rate driving, dual gate pixel architecture, and the small-area GIP block without requiring additional mask processes or changes in OLED materials. By using the DRD driving method and reducing the number of source driver ICs to half, specifically to 8, and lowering system cost, while applying a small-area GIP structure, it was possible to ensure that the panel bezel did not increase compared to conventional OLED panels, despite the increase in gate lines to 4,320. No degradation in display uniformity, motion artifacts, flicker, or signal integrity, even under 165 Hz high-refresh rate driving. Panel level tests included electrical measurement, optical integrity characterization, and reliability assessment across temperature and aging stress tests.

These results confirm that combining double-rate driving with small-area GIP is an effective approach to improving both cost and form factor in large OLED TVs. Furthermore, the proposed DRD architecture is scalable to higher resolutions and frame rates beyond 165 Hz. Reducing the number of data driver ICs also lowers system complexity and improves power efficiency, while the compact GIP structure enables thinner bezel designs for next-generation premium OLED TVs. Table 1 shows the characteristics of the fabricated 65-inch 4K OLED display panels.

4. Conclusion

We have developed a high-refresh-rate and large-size OLED display architecture combining double-rate driving, a dual-gate pixel structure, and a small-area GIP design. The proposed technologies significantly reduce the number of source driver ICs, enable stable 165 Hz operation, improve pixel charging performance, and minimize panel bezel width. Successful implementation on a 65-inch 4K OLED panel demonstrates strong feasibility for next-generation premium TV applications.

Table 1. Characteristics of the developed OLED displays for premium TVs

Item	Content	Unit
Panel size	65	Inch
Resolution	3840×2160	-
Pixel Architecture	Dual Gate	-
Panel Bezel (L/R)	4.3	mm
Frame Rate	165	Hz
Brightness	4500/450	cd/m ²
MPRT	< 4.94	ms

5. References

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